

### 01\_ML\_CIEM

- 2-D case of regular wave run-up on a beach, validated against physical model tests [Scandura & Foti, 2011]. Coupling with SWASH model by means of multi-layered (ML) piston (one-way offline coupling). [Video](#)

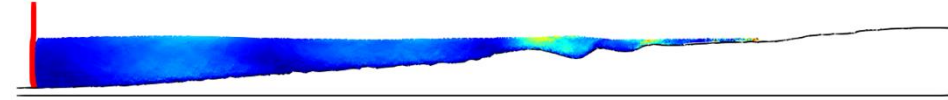
Time: 2.50 s



Time: 5.00 s



Time: 7.89 s



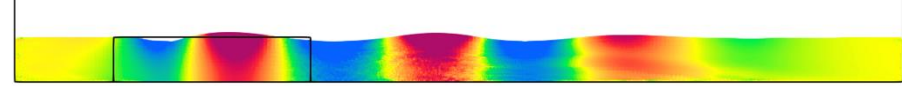
### 02\_RZ\_REGULARWAVES

- 2-D regular waves generated by means of relaxation zone. [Video](#)

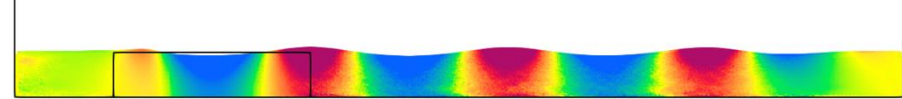
Time: 1.50 s



Time: 5.00 s



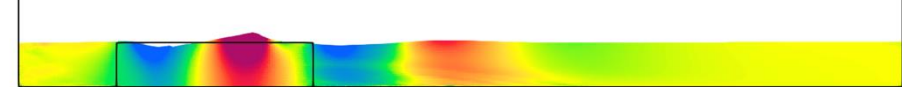
Time: 20.00 s



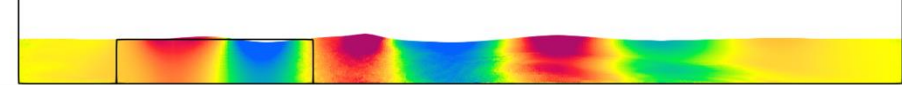
### 03\_RZ\_IRREGULARWAVES

- 2-D irregular waves generated by means of relaxation zone. [Video](#)

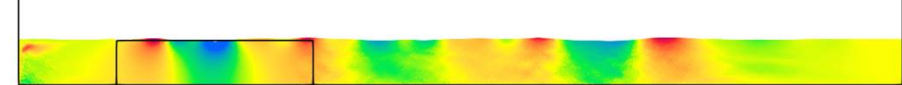
Time: 2.50 s



Time: 5.00 s

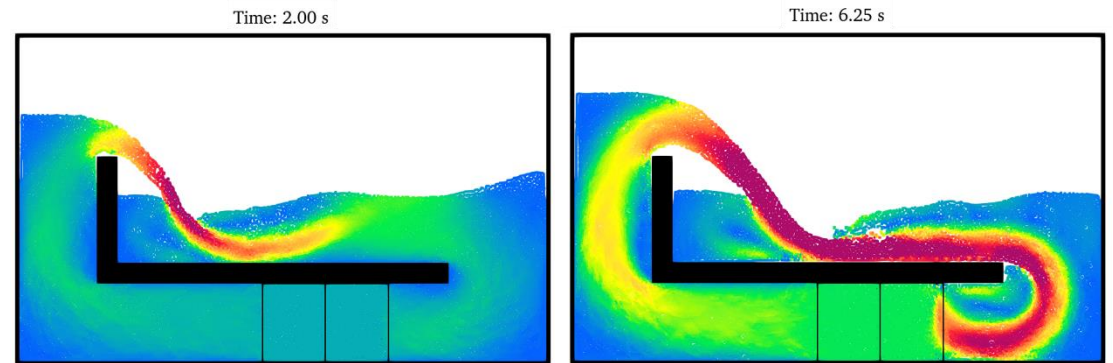


Time: 19.90 s



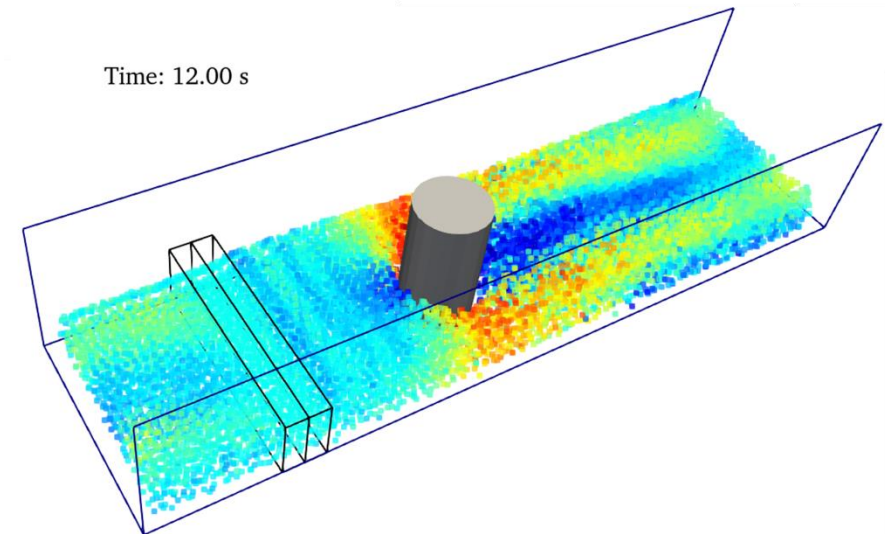
#### 04\_RZ\_FLOW2D

- 2-D flow (current) generated by means of relaxation zone by applying a uniform velocity profile, variable in time. [Video](#)



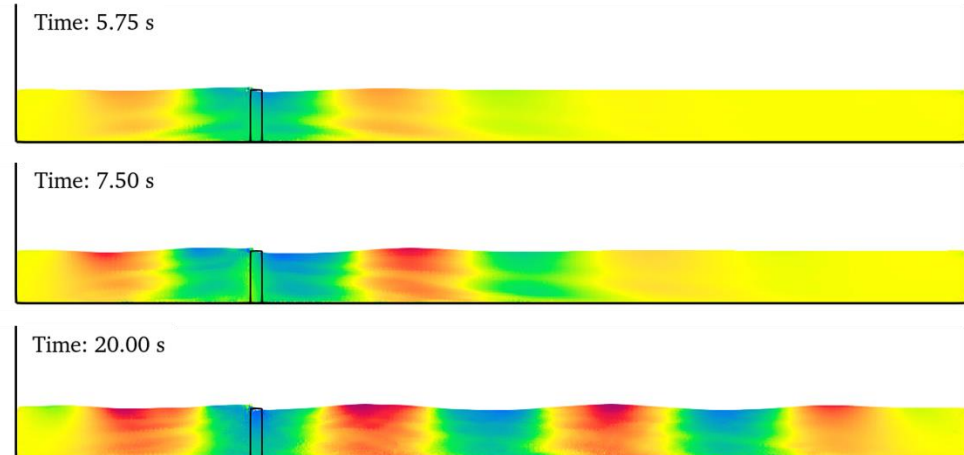
#### 05\_RZ\_FLOWCYLINDER3D

- 3-D flow (current) around a cylinder generated by means of relaxation zone by applying a uniform velocity profile, variable in time. [Video](#)



#### 06\_RZ\_COUPLING

- 2-D regular waves where the wave orbital velocities are provided by SWASH model and imposed to the SPH fluid using the relaxation zone. [Video](#)



## 07\_INLET\_AWAS\_COUPLING

- 2-D irregular waves generated by means of Open Boundary Conditions (i.e. inlet), where the imposed water surface elevation and velocity field at the inlet is provided by SWASH model. Reference: [Lowe et al. \(2022\)](#). [Video](#)

